

## MATH 230-2 Worksheet 2

Tuesday, April 15th

**Problem 1.** In this problem we outline a method for evaluating the integral

$$I = \int_{-\infty}^{\infty} e^{-x^2} dx.$$

The graph of  $e^{-x^2}$  is a bell-shaped curve. This integral appears frequently in probability and statistics (an appropriate scaling of this function is the probability density function of a normal distribution). Interestingly,  $e^{-x^2}$  has no elementary antiderivative, so traditional integration techniques will not be helpful here.

The trick to evaluating this integral is to add another dimension to the problem and then evaluate the corresponding double integral in two different ways. Let  $S_R = [-R, R] \times [-R, R]$  be the square with side lengths  $2R$  centered at the origin, and let  $D_R$  be the disk of radius  $R$  centered at the origin.

(a) Show that

$$\iint_{S_R} e^{-(x^2+y^2)} dA = \left( \int_{-R}^R e^{-x^2} dx \right)^2.$$

(b) Show that

$$\iint_{D_R} e^{-(x^2+y^2)} dA = \pi(1 - e^{-R^2}).$$

(Hint: use polar coordinates!)

(c) Taking the limit as  $R \rightarrow \infty$ , it can be shown that

$$\lim_{R \rightarrow \infty} \iint_{S_R} e^{-(x^2+y^2)} dA = \iint_{\mathbb{R}^2} e^{-(x^2+y^2)} dA = \lim_{R \rightarrow \infty} \iint_{D_R} e^{-(x^2+y^2)} dA.$$

Use this to show

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}.$$

**Problem 2.** Suppose that, in a map of a city, the city center is placed at the origin. The population density of the city is given by  $\delta(x, y) = 2000(x^2 + y^2)^{-0.2}$  people per square kilometer. How many people live within 4 km of the city center? (Use a calculator to round to the nearest whole person.)

**Problem 3.** Suppose a triangle has vertices  $(0, 0)$ ,  $(b, 0)$ , and  $(a, h)$  where  $a, b, h$  are positive and  $a < b$ . Show that the centroid of the triangle is  $(\bar{x}, \bar{y}) = (\frac{a+b}{3}, \frac{h}{3})$ .

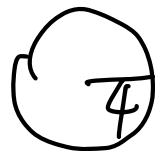
Problem 1:

$$\begin{aligned} (a) \quad \iint_{S_R} e^{-(x^2+y^2)} dA &= \int_{-R}^R \int_{-R}^R e^{-(x^2+y^2)} dx dy \\ &= \left( \int_{-R}^R e^{-x^2} dx \right) \left( \int_{-R}^R e^{-y^2} dy \right) \\ &= \left( \int_{-R}^R e^{-x^2} dx \right) \left( \int_{-R}^R e^{-x^2} dx \right) = \left( \int_{-R}^R e^{-x^2} dx \right)^2 \end{aligned}$$

$$\begin{aligned} (b) \quad \iint_{D_R} e^{-(x^2+y^2)} dA &\stackrel{\substack{x=r\cos\theta \\ y=r\sin\theta}}{=} \int_0^{2\pi} \int_0^R e^{-r^2} r dr d\theta \\ &= \int_0^{2\pi} \left( -\frac{1}{2} e^{-r^2} \Big|_{r=0}^{r=R} \right) d\theta \\ &= \int_0^{2\pi} \left( -\frac{1}{2} e^{-R^2} - \left( -\frac{1}{2} e^{-0^2} \right) \right) d\theta \\ &= \pi (1 - e^{-R^2}). \end{aligned}$$

$$\begin{aligned} (c) \quad \lim_{R \rightarrow \infty} \iint_{S_R} &= \iint_{\mathbb{R}^2} = \lim_{R \rightarrow \infty} \iint_{D_R} \\ &\quad \downarrow \text{by (a)} \qquad \qquad \qquad \downarrow \text{by (b)} \\ \lim_{R \rightarrow \infty} \left( \int_{-R}^R e^{-x^2} dx \right)^2 &= \lim_{R \rightarrow \infty} \pi (1 - e^{-R^2}) = \pi \\ \Rightarrow \int_{-\infty}^{\infty} e^{-x^2} dx &= \sqrt{\pi}. \end{aligned}$$

Problem 2:



$$\iint_D f(x, y) dA$$

$$D = \{x^2 + y^2 \leq 4^2\}$$

$$(\quad = \iint_{\{x^2 + y^2 \leq 4^2\}} 2000 (x^2 + y^2)^{-0.2} dA)$$

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$\int_0^{2\pi} \int_0^4 2000 (r^2)^{-0.2} \cdot r dr d\theta$$

$$= \int_0^{2\pi} \int_0^4 2000 \cdot r^{0.6} dr d\theta$$

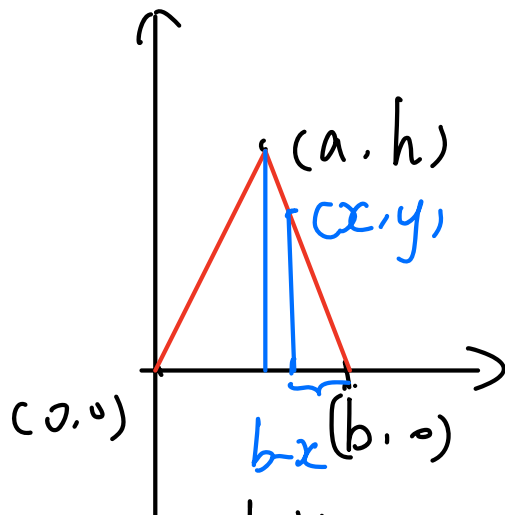
$$= 2000 \cdot 2\pi \cdot \int_0^4 r^{0.6} dr$$

$$= 2000 \cdot 2\pi \cdot \left( \frac{1}{1+0.6} r^{1+0.6} \right) \Big|_0^4$$

$$= 4000\pi \cdot \frac{1}{1.6} \cdot 4^{1.6} \approx 72174$$

# Problem 3

$$\bar{x} = \frac{\iint_R x \, dA}{\iint_R 1 \, dA}$$



$$\iint_R x \, dA = \int_0^a \int_0^{\frac{h}{a}x} x \, dy \, dx + \int_a^b \int_0^{\frac{b-x}{b-a} \cdot h} x \, dy \, dx$$

$$\frac{b-x}{b-a} = \frac{y}{h} \Rightarrow y = \frac{b-x}{b-a} \cdot h$$

$$= \int_0^a x \cdot \frac{h}{a} x \, dx + \int_a^b x \cdot \frac{b-x}{b-a} \cdot h \, dx$$

$$= \frac{h}{a} \cdot \frac{1}{3} a^3 + \frac{h}{b-a} \left( \frac{1}{2} x^2 b - \frac{1}{3} x^3 \right) \Big|_a^b$$

$$= \frac{h}{3} a^2 + \frac{h}{b-a} \left[ \frac{1}{2} b^3 - \frac{1}{3} b^3 - \left( \frac{1}{2} a^2 b - \frac{1}{3} a^3 \right) \right]$$

$$= \frac{h}{3(b-a)} (a^2 b - a^3) + \frac{3h}{3(b-a)} \left( \frac{1}{6} b^3 - \frac{1}{2} a^2 b + \frac{1}{3} a^3 \right)$$

$$= \frac{\frac{1}{2} h b (b^2 - a^2)}{3(b-a)} = \frac{1}{6} h b (b+a)$$

$$\Rightarrow \bar{x} = \frac{\iint_R x \, dA}{\iint_R 1 \, dA} = \frac{\frac{1}{6} h b (b+a)}{\frac{1}{2} b \cdot h} = \frac{a+b}{3}$$

$$\begin{aligned} \iint_R y \, dA &= \int_0^a \int_0^{\frac{h}{a}x} y \, dy \, dx + \int_a^b \int_0^{\frac{b-x}{b-a} \cdot h} y \, dy \, dx \\ &= \int_0^a \frac{1}{2} \left( \frac{h}{a}x \right)^2 dx + \int_a^b \frac{1}{2} \left( \frac{b-x}{b-a} \cdot h \right)^2 dx \\ &= \frac{h^2}{2a^2} \cdot \frac{1}{3} a^3 + \frac{h^2}{2(b-a)^2} \cdot \left( \frac{1}{3} (x-b)^3 \right) \Big|_a^b \\ &= \frac{1}{6} h^2 a + \frac{h^2}{6(b-a)^2} [-(a-b)^3] \\ &= \frac{1}{6} h^2 a - \frac{1}{6} h^2 (a-b) = \frac{1}{6} h^2 b \end{aligned}$$

$$\Rightarrow \bar{y} = \frac{\iint_R y \, dA}{\iint_R 1 \, dA} = \frac{\frac{1}{6} h^2 b}{\frac{1}{2} b \cdot h} = \frac{h}{3}$$

(i.e.,  $\delta(x,y)$  is a constant!)

In general, the centroid of the  $\Delta$  with vertices  $(x_1, y_1), (x_2, y_2), (x_3, y_3)$  is given by  $\bar{x} = \frac{x_1 + x_2 + x_3}{3}, \bar{y} = \frac{y_1 + y_2 + y_3}{3}.$